

1. Electric current is the flow of electric charge through a conductor.
2. amperes (A).
3. I
4. A circuit needs a closed (complete) path for current to flow.
5. The battery provides the energy (voltage) needed to push electric current through the circuit.
6. A conductor is a material that allows electric current to flow easily.
7. An insulator is a material that does not allow electric current to flow easily.
8. Voltage
9. Electric current heats the wire when it flows through it.
10. A switch controls the flow of current by opening or closing the circuit.
11. Electric current flows in a circuit when the circuit is closed, allowing charges to move from the positive terminal of the battery, through the components, and back to the negative terminal. The battery provides the energy needed to push these charges through the circuit.
12. Conductors allow electric current to flow easily because they have many free electrons, with metals like copper being common examples. Insulators, on the other hand, do not allow current to pass through as they have very few free electrons; materials like rubber and plastic are good insulators.
13. A fuse protects an electrical circuit by breaking the flow of current when it becomes too high. It contains a thin wire that melts when excessive current passes through, preventing damage to appliances and reducing the risk of fire.
14. Ohm's law states that the current flowing through a conductor is directly proportional to the voltage across it, provided the temperature remains constant. Formula: $V=I \times R$
15. In a series circuit, all components are connected one after another in a single path, so the same current flows through each component. In a parallel circuit, components are connected in multiple branches, allowing current to divide and flow through each branch separately, so each branch gets the same voltage.
16. Electric current produces a magnetic field around the wire, which can magnetize nearby materials or create an electromagnet.
17. Safety precautions while handling electricity:
 - Never touch electrical appliances or wires with wet hands.
 - Avoid using damaged cords or plugs.

- Switch off the power before repairing or installing electrical devices.
- Do not overload sockets.
- Keep electrical devices away from water.
- Use insulated tools and wear protective gear when necessary.

18. Resistors limit the flow of electric current in a circuit and help control voltage across components.

19. As the temperature of a conductor increases, its electrical resistance also increases. This happens because the atoms in the conductor vibrate more at higher temperatures, causing more collisions for the electrons moving through it. Hence, the flow of current becomes slightly harder, leading to higher resistance.

20. An electric bell works on the principle of electromagnetism. When an electric current passes through a coil of wire, it behaves like a magnet. This temporary magnetism is used to repeatedly move a hammer that strikes the bell. When the switch is pressed, current flows through the electromagnet. The soft iron core becomes magnetized and attracts the nearby iron armature. As the armature moves towards the electromagnet, the hammer attached to it strikes the bell and produces sound. At the same time, this movement breaks the circuit at the contact point, stopping the current. Once the current stops, the electromagnet loses its magnetism, and the armature returns to its original position due to a spring. The circuit gets completed again, current flows, and the cycle repeats rapidly. This rapid making and breaking of the circuit causes the hammer to strike the bell repeatedly, producing a continuous ringing sound.

21. A simple electric circuit consists of a source of electricity (like a cell or battery), connecting wires, a switch, and a load such as a bulb. The cell provides the necessary electrical energy. The wires connect all components, forming a closed path for current to flow. The switch is used to open or close the circuit, and the bulb uses the electrical energy to produce light. When the switch is closed (ON), the circuit becomes a complete closed path. This allows electric current to flow from the positive terminal of the cell, through the wires, to the bulb. The filament inside the bulb heats up due to the current, causing it to glow. The current then returns to the negative terminal of the cell, completing the circuit. If the switch is open (OFF), the path is broken. No current flows, and the bulb does not glow. Thus, a simple electric circuit works by providing a continuous path for the flow of current through its components.

22. Electric current plays a vital role in our everyday activities and is used in many practical applications that make life comfortable and efficient. One of the most common uses is lighting. Electric bulbs, tube lights, and LED lamps work using electric current, helping us see clearly in homes, schools, and streets. Another major application is in household appliances. Devices such as fans, refrigerators, mixers, washing machines, heaters, and air conditioners all operate using electric current. These appliances make cooking, cleaning, cooling, and heating tasks much easier. Electric current is also widely used in communication systems. Mobile phones, computers, televisions, and radios all require electricity to function. Similarly, electric current powers transportation systems such as electric trains, trams, and electric vehicles (EVs), making travel faster and eco-friendly. In addition, electricity is used in medical equipment like X-ray machines, ECG monitors, and life-support devices, which are essential for diagnosis and treatment. It is also used in industries to run machines, motors, and tools for manufacturing. Thus, electric current has numerous applications in lighting, household appliances, communication, transportation, healthcare, and industries, making it an indispensable part of modern life.
23. An electromagnet is a type of temporary magnet created by passing electric current through a coil of wire. It works on the principle of electromagnetism, which states that an electric current produces a magnetic field. When a coil of insulated wire is wrapped around a soft iron core and connected to a battery, the current flowing through the coil magnetizes the iron core. This turns it into a strong magnet. The magnetism lasts only as long as the current flows; when the current stops, the magnetic field disappears. The strength of an electromagnet can be increased by adding more turns to the coil or increasing the current. Electromagnets are widely used because they can be turned on and off easily. They are used in electric bells, relays, and telephones for switching purposes. In cranes, large electromagnets lift and move heavy objects like scrap iron in junkyards. Electromagnets are also used in motors and generators, where they help convert electrical energy into mechanical energy and vice versa. They are found in loudspeakers, where they help produce sound, and in MRI machines in hospitals for medical imaging. Thus, electromagnets are essential in many devices and machines due to their controllable and powerful magnetic properties.
24. Electricity is one of the most important resources in modern society and plays a central role in almost every aspect of daily life. It powers our homes, industries, hospitals, schools, transportation systems, and communication networks, making modern living

comfortable, efficient, and productive. In households, electricity is essential for lighting, cooking, cooling, heating, and running appliances like refrigerators, washing machines, computers, and televisions. Without electricity, most daily activities would become difficult and time-consuming. In education and communication, electricity powers computers, projectors, the internet, mobile phones, and televisions. These tools help students learn effectively and allow people to stay connected across the world. In healthcare, electricity is crucial for operating medical equipment such as X-ray machines, ventilators, ECG monitors, and life-support systems. Modern hospitals depend heavily on electricity to provide accurate diagnosis and emergency treatment. Electricity also plays a major role in industries and economic development. Machines, motors, and manufacturing units all require electricity. It increases productivity, supports employment, and contributes to a country's growth. In transportation, electric trains, metro systems, and electric vehicles promote faster travel and reduce pollution, making transportation more efficient and eco-friendly. Overall, electricity has transformed modern society by improving living standards, enhancing communication, supporting healthcare, driving industries, and promoting sustainable development. Its availability and proper use are essential for progress and a better quality of life.

25. Renewable energy sources are natural resources that can be replenished continuously and will not run out over time. Common renewable sources include solar energy, wind energy, hydropower, biomass, and geothermal energy. Unlike fossil fuels such as coal and oil, these sources are clean, sustainable, and environmentally friendly. These energy sources play a major role in generating electric current. Solar panels convert sunlight directly into electricity using photovoltaic cells. Wind turbines use the kinetic energy of wind to rotate blades connected to generators, producing electric current. Hydroelectric power plants use the force of flowing or falling water to spin turbines and generate electricity. Biomass can be burned to produce steam that runs turbines, while geothermal energy uses heat from inside the Earth to produce electricity. The significance of renewable energy is very high in today's world. They help reduce air pollution, decrease greenhouse gas emissions, and slow down climate change. Renewable sources also provide a long-term, reliable supply of electricity and reduce dependence on non-renewable fuels. They promote sustainable development and help meet the growing global demand for energy in an eco-friendly way. Thus, renewable energy sources are essential for generating electricity in a clean, sustainable, and environmentally responsible manner.